Gloved Hands, Blind Models: Bridging the Domain Gap in Surgical Hand Pose Estimation

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AIM G1
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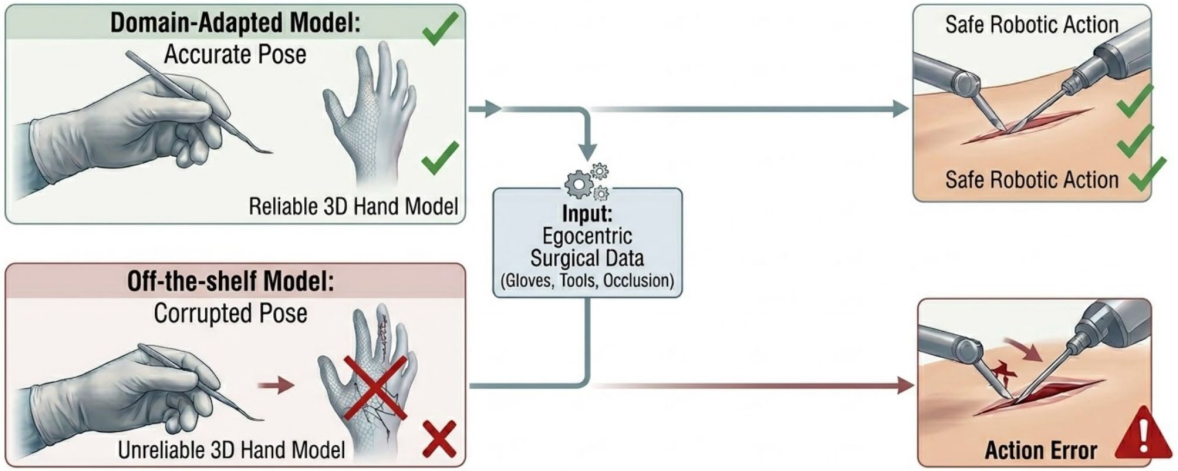
Why surgical hand tracking matters

- Surgical AI (VLA models) learns actions from hand motion
- Hands are represented as a **102D MANO** action space
- If hand tracking fails, the downstream system cannot learn reliably

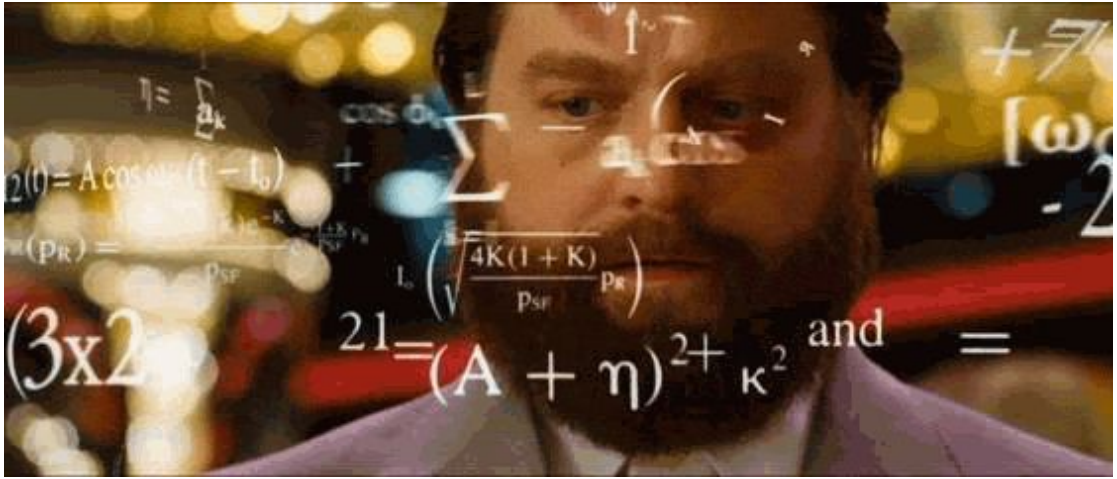


Off-the-shelf models fail on surgical data

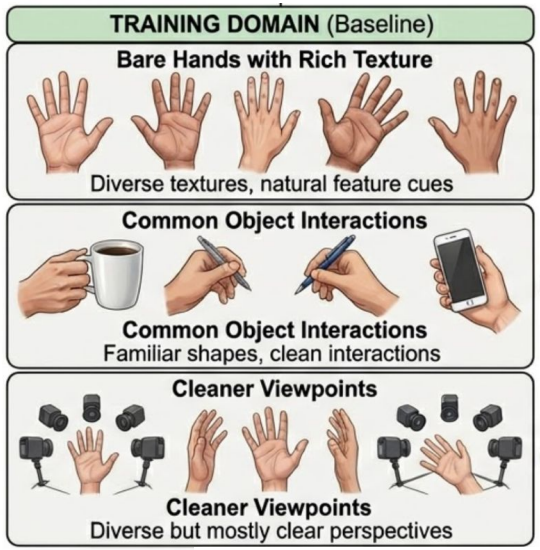
So we shifted from building VLAs to fixing the hand-tracking bottleneck in surgical settings

What are we trying to understand?

- How severe is the performance gap?
- Where do models fail: detection, position, or shape?
- Can domain adaptation fix it?



Why surgery is fundamentally different



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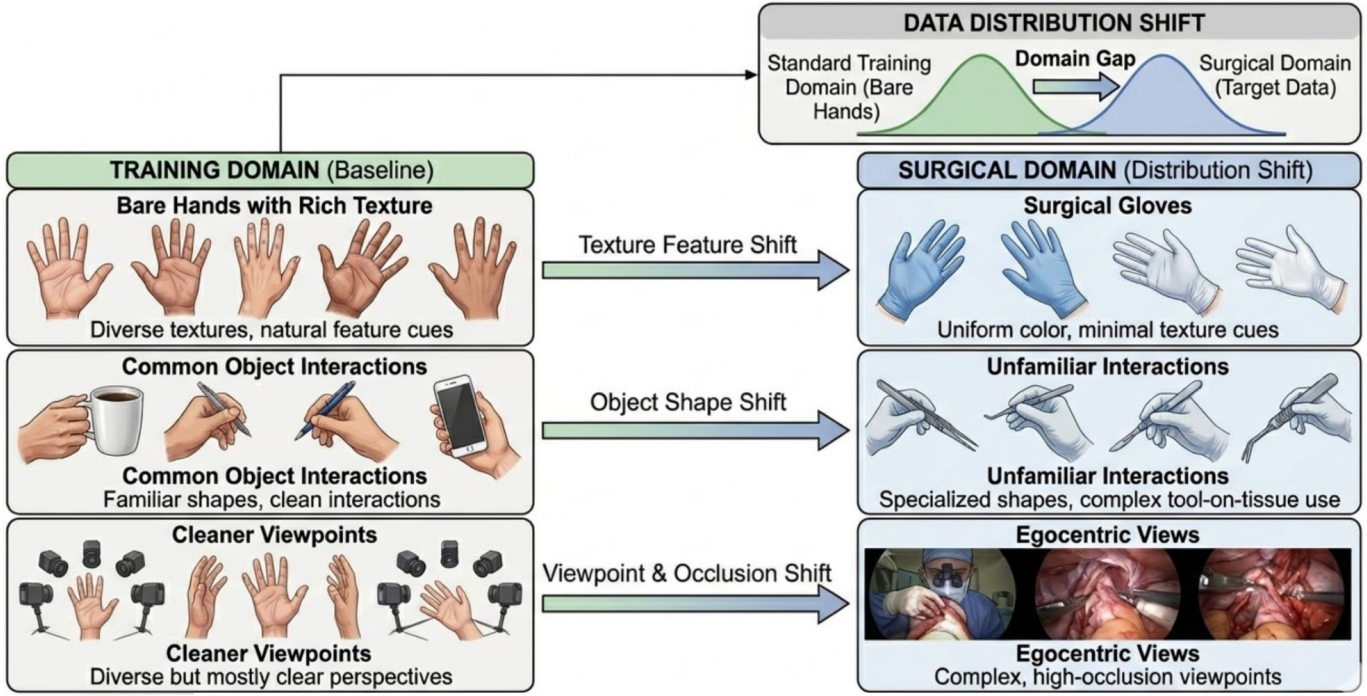
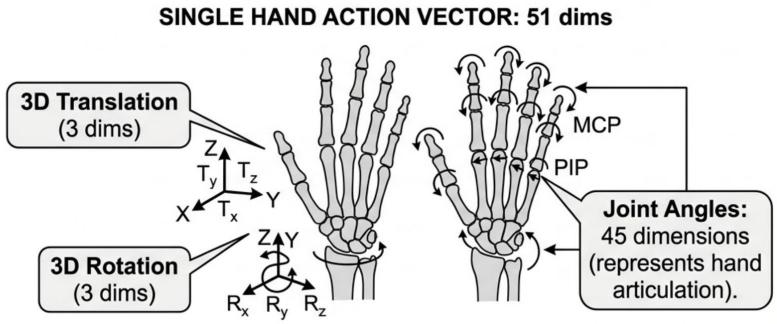


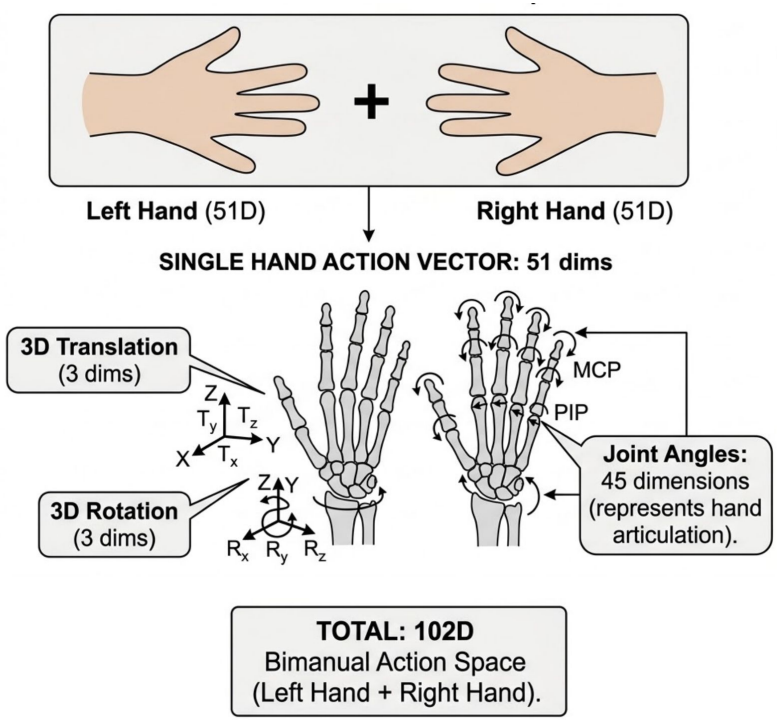
How hand motion becomes the action space

- VLA models represents each hand with **51 action dimensions**
- These include **3D translation, 3D rotation, and 45 joint angles**



How hand motion becomes the action space

- VLA models represents each hand with **51 action dimensions**
- These include **3D translation, 3D rotation, and 45 joint angles**
- Two hands together form a **102D bimanual action space**
- MANO is used as the hand model for joint structure, but **shape pa are not part of the action vector**



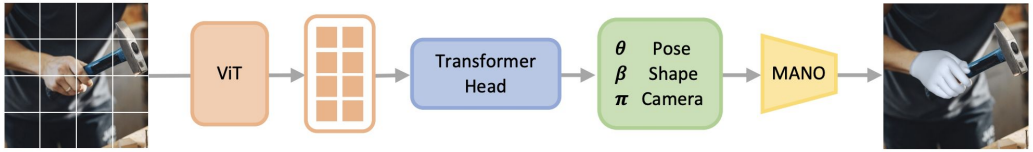
Key takeaway

Reliable MANO estimation is essential for surgical action learning

Two strong hand models

HaMeR (CVPR 2024)

- Fully transformer-based
- ViT-H backbone + transformer decoder
- Direct MANO + camera regression
- Single-stage hand mesh recovery
- Larger model, no detector



hand image → transformer reads image → predicts hand pose + hand shape + camera → MANO generates final 3D hand.

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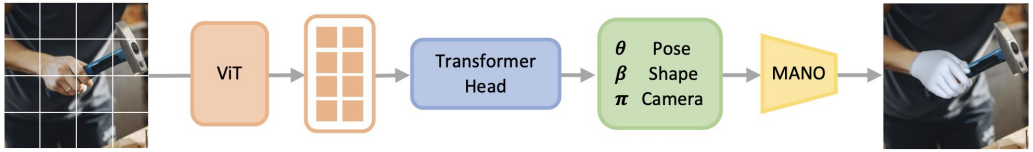
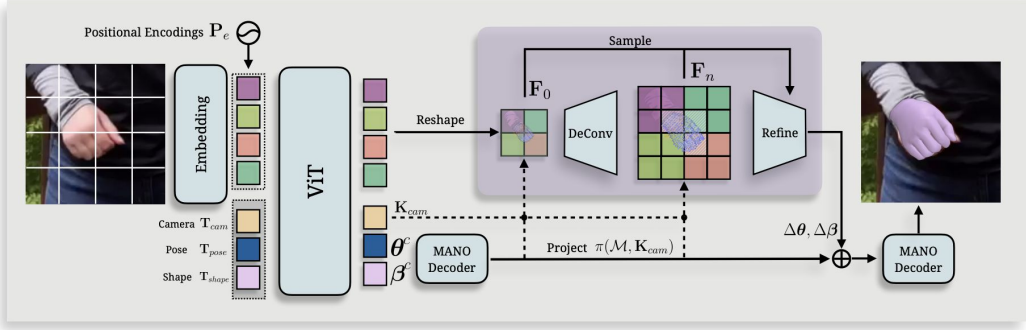
WiLoR (CVPR 2025)

- Full pipeline: detection + reconstruction
- Real-time hand detector with handedness
- ViT-based MANO regression
- Coarse-to-fine refinement module
- Better image alignment for final pose





full image → detector → hand crop → ViT rough MANO estimate → refinement module → final MANO estimate.

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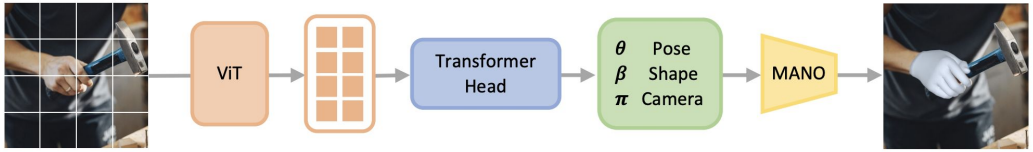
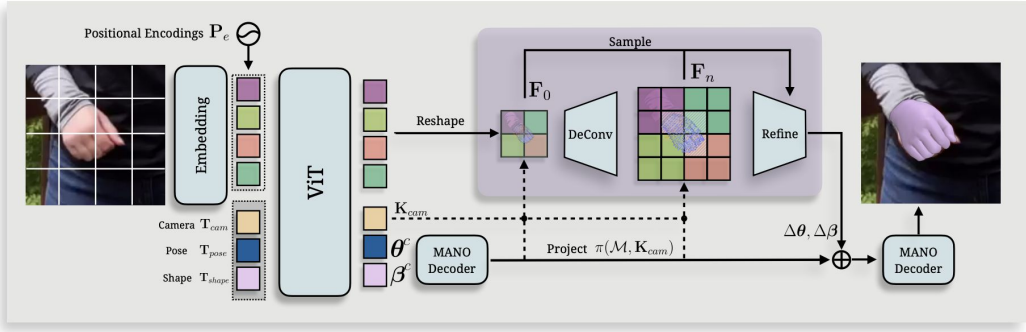
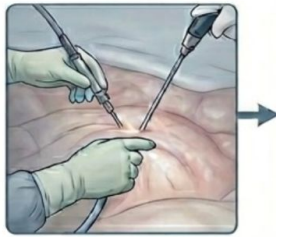




Key takeaway

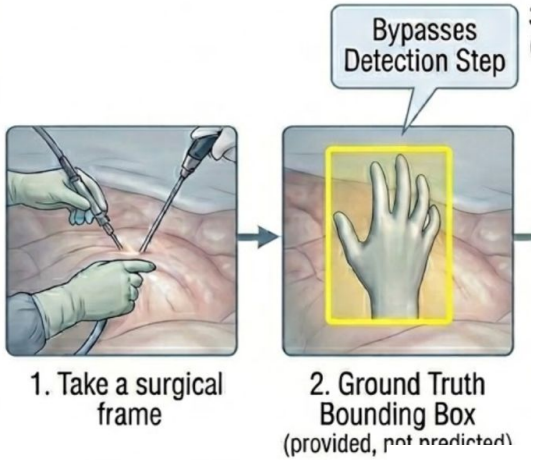
- **Both are strong on standard hand-pose benchmarks**
- **Neither had been tested on surgical data before this project**

How we evaluate the models



1. Take a surgical frame

How we evaluate the models



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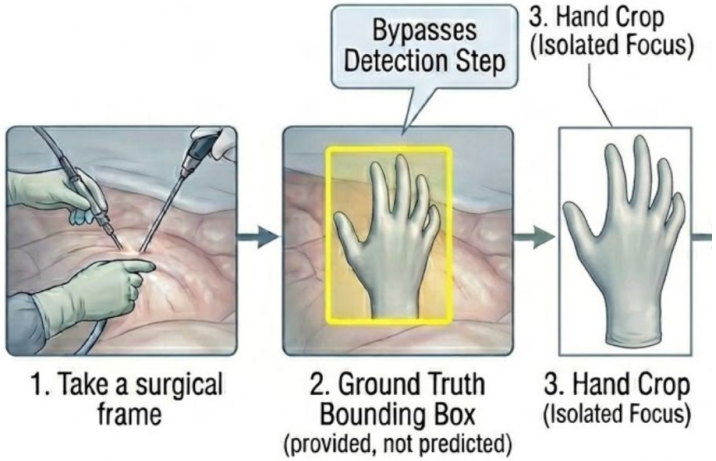

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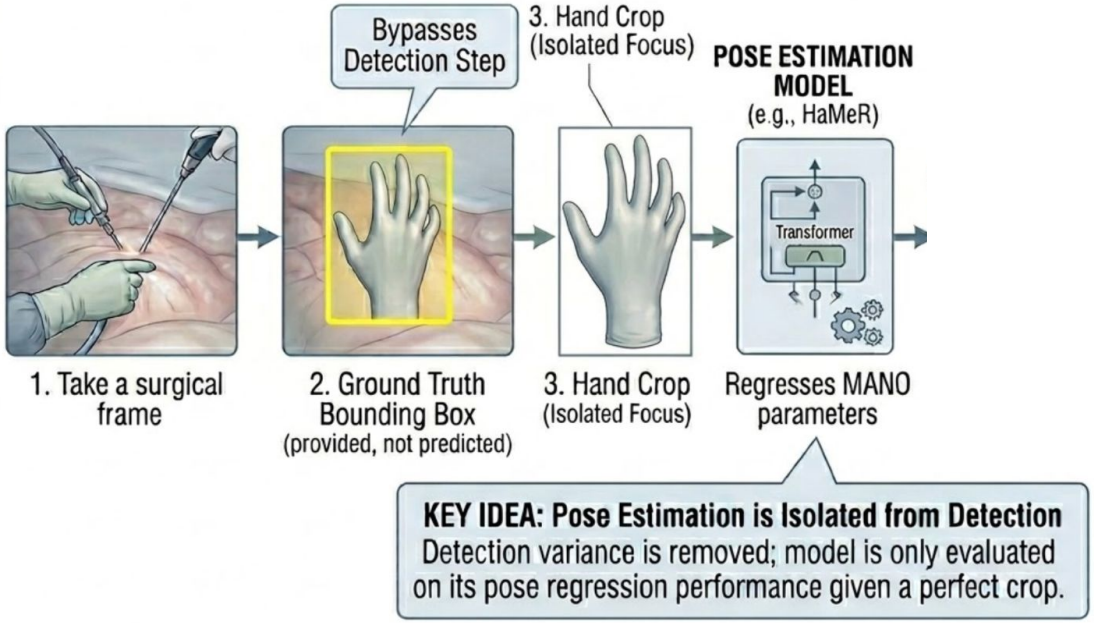

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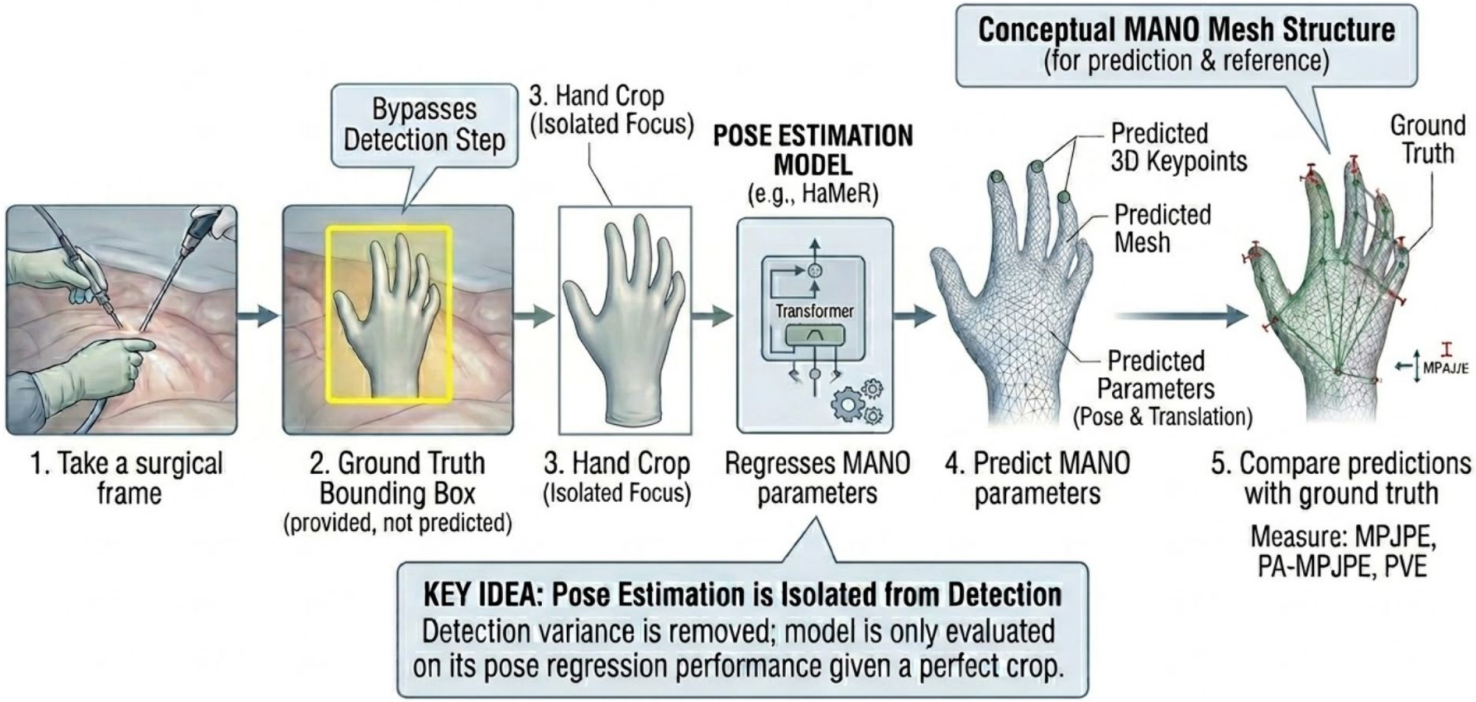

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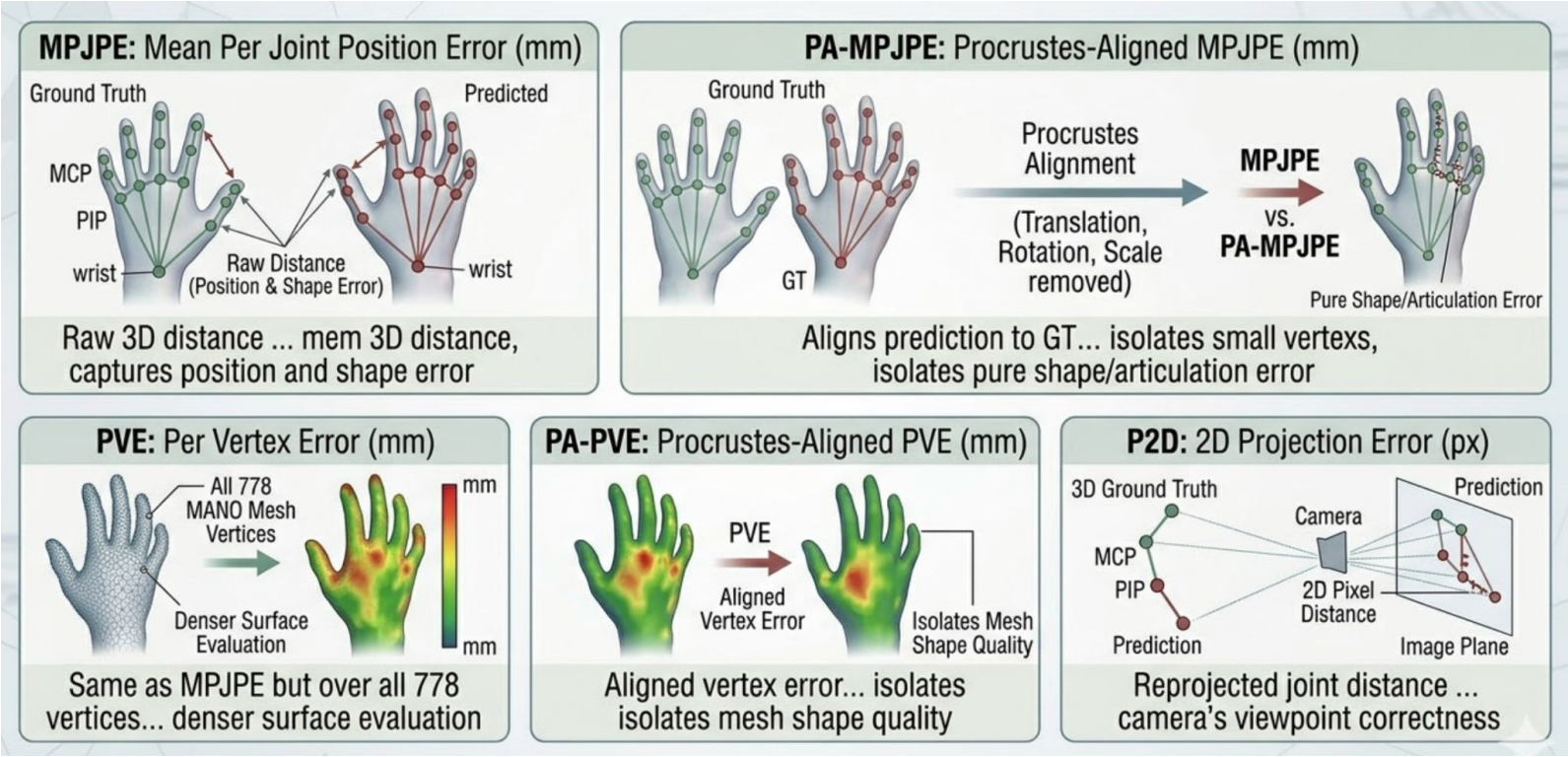
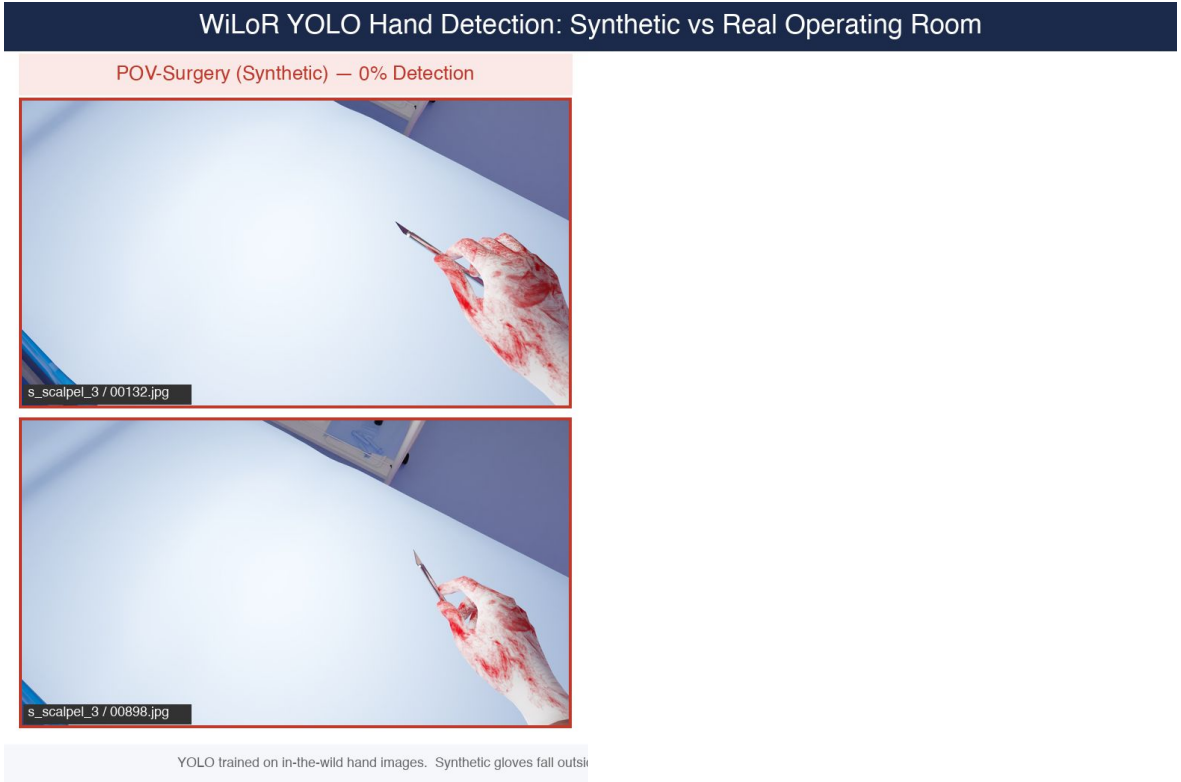


Baseline Results: Detection Failure

First failure: detection

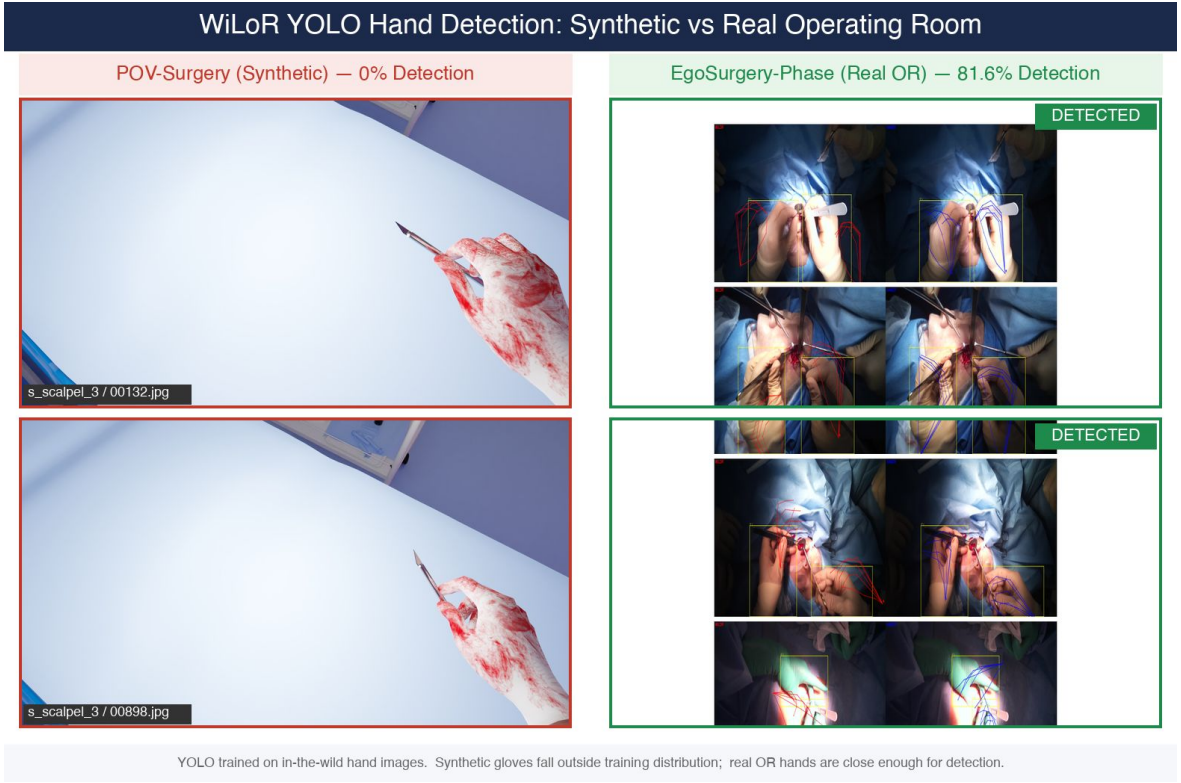
- Synthetic (POV-Surgery):
  - 0% detection



The same detector behaves very differently across domains

First failure: detection

- Synthetic (POV-Surgery):
  - 0% detection
- Real OR (EgoSurgery):
  - 81.6% detection
  - (115 / 141 frames)



The same detector behaves very differently across domains

Core result: both models hit the same ceiling

Model	MPJPE	PA-MPJPE	PVE	PA-PVE	P2D
WiLoR	50.36 mm	10.69 mm	47.92 mm	10.01 mm	26.0 px
HaMeR	54.31 mm	11.17 mm	51.68 mm	10.48 mm	30.58 px

- **Different models converge to the same ~11 mm PA-MPJPE ceiling**
- **That points to a domain-gap limit, not a model-capacity limit**

Where does the error come from?

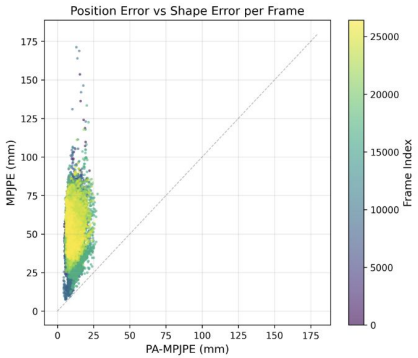
- Global position error: ~79%
- Shape / articulation error: ~21%

Example:

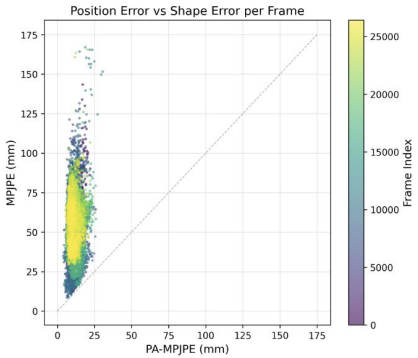
- HaMeR: 54.3 mm MPJPE vs 11.2 mm PA-MPJPE

Interpretation:

- The models capture hand structure reasonably well
- But they place the hand at the wrong 3D location



(a) WiLoR: MPJPE vs. PA-MPJPE per frame.



(b) HaMeR: MPJPE vs. PA-MPJPE per frame.

Detection varies by surgical phase

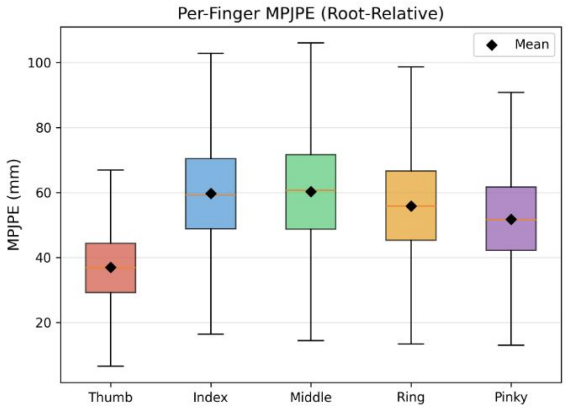
Phase	Detection
Incision	90%
Dissection	77%
Closure	87%

- **Performance changes with environment and scene difficulty, especially occlusion and lighting**

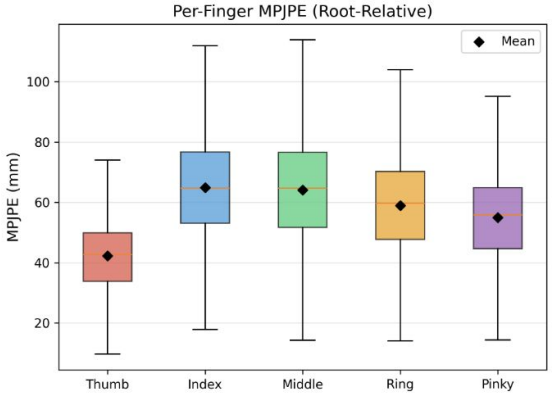
Which fingers are hardest?

Finger	WiLoR	HaMeR
Thumb	36.93	42.24
Index	59.73	64.90
Middle	60.23	64.13
Ring	55.83	58.96
Pinky	51.67	54.92

- **The index finger has the highest error**
- **Likely because it is most active and most occluded during tool use**



(a) WiLoR per-finger MPJPE distribution.



(b) HaMeR per-finger MPJPE distribution.

What did we learn?

- Both models converge to the same ~11 mm ceiling
- Detection and regression fail for different reasons
- Most of the error comes from position, not hand shape

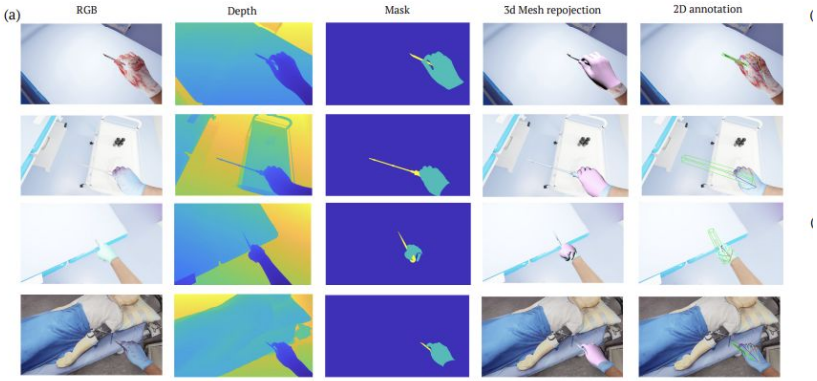
Key challenges we solved

Challenge	Solution
Complete detection failure. YOLO detects 0% of gloved surgical hands	Designed a crop-regress protocol to evaluate regression independently from detection
Platform incompatibility. HaMeR dependencies break on Python 3.13 and macOS	Installed minimal inference dependencies and forced CPU for float64 operations
Coordinate space mismatch. Predictions and annotations use different camera frames	Implemented full coordinate conversion for fair evaluation
Project pivot. Hand tracking failure blocked downstream VLA work	Refocused the project on systematic hand pose evaluation and domain adaptation

How do we fix it?

- Fine-tune on **POV-Surgery training data (~70K frames)**
- Keep the same model and evaluation pipeline
- Test whether supervised adaptation closes the gap

- **Reduce PA-MPJPE from ~11 mm**



Phase 1: Complete

Off-the-shelf baselines on POV-Surgery. Established the ~11mm domain gap ceiling.

Phase 2: Complete

Qualitative real OR evaluation. Discovered the asymmetric domain gap.

Phase 3: Active

Fine-tune WiLoR on POV-Surgery. Execute data efficiency and backbone freezing experiments.

Phase 4: Upcoming

Error taxonomy and cross-domain transfer analysis (testing the synthetic-tuned model on real OR footage).

Feedback we'd value

Our error decomposition shows 79% of total error is global position and only 21% is hand shape the **models understand hand structure but misplace it in 3D**. For fine-tuning, **should we freeze the ViT backbone** and only adapt the regression head, **or fine-tune end-to-end** at the risk of catastrophic forgetting on pov-surgical dataset?

- Will the synthetic data improve the accuracy?
- Any other pipeline we can try?

Presentation Delivery: Team Contributions

Ankit Pal (Aadi)

- WiLoR evaluation pipeline
- Detection analysis
- Error decomposition
- Cross-domain study
- Writing and presentation

Yuchang Su

- HaMeR evaluation pipeline
- Coordinate alignment
- Visualization tools
- Environment setup and debugging

Thank you!!
Question?

Real-world performance: asymmetric domain gap

- Synthetic detection: 0%
- Real OR detection: 81.6%
- Bimanual detection: 53.9% (76 / 141 frames)
- Total detected hands: 191

- **Detection works much better on real OR footage than synthetic**
- **Pose estimation still struggles in both settings**

How far are we from adapted surgical performance?

- Standard benchmarks: roughly 5-7 mm
- Our zero-shot surgical results: roughly 11 mm
- Fine-tuned surgical baselines: roughly 4.3 mm

- **There is a clear gap between zero-shot transfer and adapted surgical performance**
- **Fine-tuning is the most direct path to closing it**